# Ideation Phase
## Project Initialization and Planning Phase

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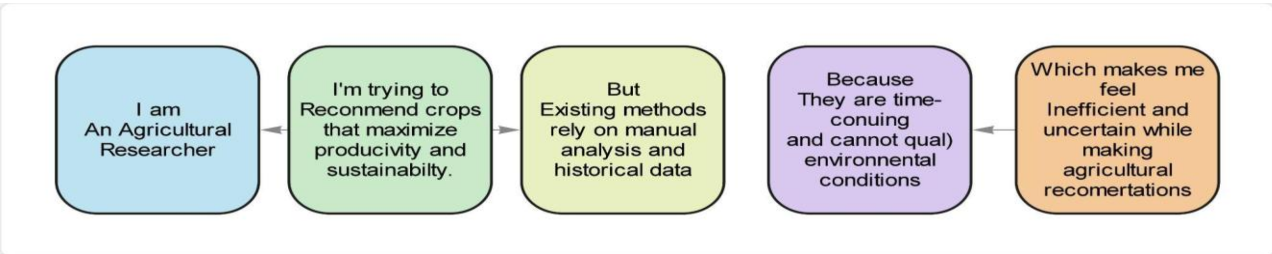
| Date | 10 July 2026 |
| Team ID | |
| Project Name | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 2 Marks |

**Customer Problem Statement:**

Farmers and agricultural stakeholders struggle to identify the most suitable crop due to variations in soil nutrients and environmental conditions. The lack of accurate, data-driven recommendations often results in reduced crop yield, inefficient resource utilization, and financial losses. OptiCrop addresses this challenge by leveraging machine learning to provide intelligent crop recommendations based on soil and climatic parameters.

**Example:**


| Problem Statement (PS) | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | A farmer | Select the most suitable crop for my farmland | I don't know which crop matches my soil and weather conditions | I lack scientific guidance and agricultural data analysis | Confused and worried about crop failure and financial loss |
| PS-2 | An agricultural researcher | Recommend crops that maximize productivity | Existing methods rely on manual analysis and historical data. | They are time-consumin g | Inefficient and uncertain while making farming decisions |